

Learned index structures

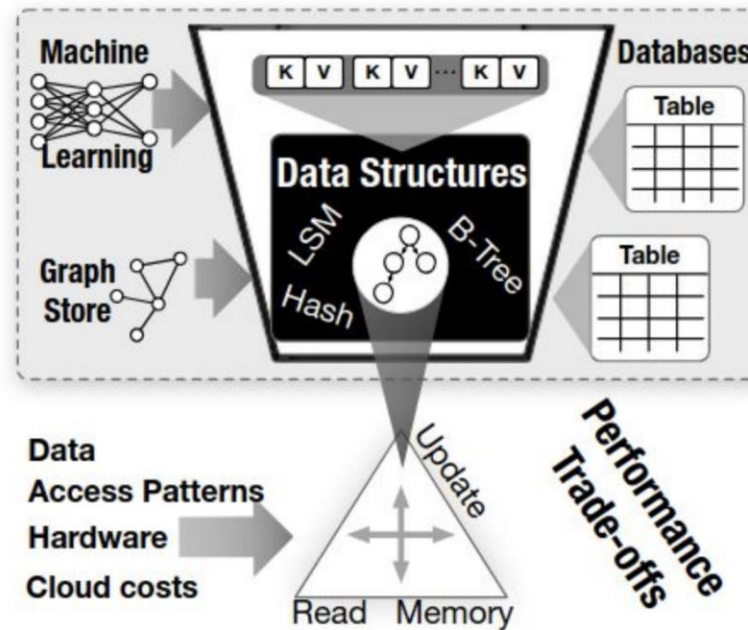
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Data structures

- Defines how data is stored (layout)
- Defines how data is accessed
- Design depends on
 - Type of data
 - Workload characteristics and requirements
 - Hardware

There is no perfect data structure

- Trade-off between performance, memory, storage
 - Read/write/memory amplification



Database index

- Used to locate data in a database
- Supported operations: point lookup, range scan, prefix scan, min/max

Common index structures

- Hash map
- B-tree / B+ -tree
- Log-Structured Merge tree (LSM)
- Trie
- Bitmap index
- Bloom filter

Why learned index?

- Currently, no good way to predict performance impact on new workloads, hardware
- Takes a long time to design and optimize a new structure

Discussion

- What are the challenges with deploying learned indices?
- How can learned indices support updates? What are the downsides/overheads?